SYSTEM REFERENCE · FOR CLIENT SIGN-OFF

Advance Deposit Tracker

Formula & Calculation Reference

Every formula, threshold and derived value in the system,
documented for joint review and confirmation

Document type: Calculation Specification

Version 2.3 · 11 July 2026

Reflects production as deployed on 11-Jul-2026 — Cost of Fund per the client-confirmed grace rule (zero within grace, retroactive from Estimated ETD once exceeded, early-shipment gains kept), the Ship Date entry rule that stops the charge (recordable even after payment lock), per-currency Cost of Fund display, and role-scoped analytics (merchandisers see own data only)

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> **Purpose of this document.** It lists every formula the platform uses so we can confirm, line by line, that the system's mathematics matches your business intent. A short companion document — the *Formula Confirmation Questionnaire* — collects the handful of points where we would like your explicit confirmation.

Conventions: all day counts are **calendar days**; *today* means the server date at calculation time; amounts are rounded to 2 decimals at the end of each formula; soft-deleted requests are excluded from every calculation and report.

1 Conventions & Configurable Parameters

Three business parameters drive the risk and cost formulas. They are stored in system configuration and editable by a Super Admin without code changes:

Parameter	Current value	Meaning & where it is used
`etd_grace_days`	**10** days	Grace period added to the supplier's Estimated ETD to form the **Grace ETD**. Drives overdue days, risk status, auto-flagging — and acts as the **threshold** for Cost of Fund: nothing is charged within grace; once exceeded, the charge counts retroactively from the Estimated ETD itself (Section 4).
`cost_of_fund_rate`	**0.12** (12% p.a.)	Annual interest rate applied in the Cost of Fund formula. Changed from 18% to 12% on 10-Jul-2026 to match the rate used throughout your sheet.
`cost_of_fund_grace_days`	30 days	**No longer used.** A leftover from the original payment-relative formula; kept in configuration only for compatibility.

Snapshot refresh cadence: every request's metrics are recomputed automatically every **30 minutes**, immediately after any payment entry/update, immediately after a request is edited, and on demand via the **Recalculate** button (Super Admin / Finance Admin).

2 Request Intake — Deposits & Numbering

2.1 Deposit amount and deposit percentage

The two fields are captured **independently** — the system does not derive one from the other on the in-app or public forms. Validation: `0 < deposit_percentage ≤ 100`; `deposit_amount > 0`; `total_supplier_invoice_amount > 0`.

```
Balance (%) shown on forms = 100 − Deposit %          (display only, not stored)
```

The one channel that auto-computes the amount is the **Google Form integration**:

```
Deposit Amount (Google Form intake) = round( Total Supplier Invoice Amount × Deposit % ÷ 100
```

2.2 Fixed deposit suppliers

If a supplier has a **fixed deposit amount** configured in the supplier master, both forms force the deposit amount to that value, clear the percentage, and lock the field. No arithmetic — a straight substitution.

2.3 Request numbering

```
Request # = ADT-{year}-{5-digit sequence}      e.g. ADT-2026-00417
sequence  = highest existing number for that year + 1   (concurrency-safe)
```

3 Shipment Risk — Grace ETD, Overdue Days, Risk Status

3.1 Grace ETD

```
Grace ETD = Estimated ETD + etd_grace_days            (currently +10 calendar days)
```

If a request has no Estimated ETD, no risk metrics can be computed and its status is *pending*.

3.2 Overdue days (the central risk metric)

```
reference date = Actual Shipment Date   (once recorded by Accounts)
                 otherwise today               (keeps counting daily)

Overdue Days = max( reference date − Grace ETD , 0 )
```

Once the actual shipment date is recorded, the value **freezes permanently** at that date. While unshipped it grows by one every day. This same number is used for the delay buckets, supplier flagging and the NPA panel. (Cost of Fund uses grace only as a threshold — when charged, the amount runs on the grace-free day count of Section 3.4, not this number.)

3.3 Risk status classification

Status	Condition (on Overdue Days)
Pending	No Estimated ETD on the request (nothing to measure against)
On Time	Overdue Days = 0 (shipped, or still within Grace ETD)
Delayed	1 – 30 days overdue
Critical	More than 30 days overdue

3.4 Actual ETD Overdue Days ALIGNED 10-JUL-2026

```
Actual ETD Overdue Days = ( Actual Shipment Date, or Today while unshipped ) − Estimated ETD
```

This is your sheet's "Actual ETD Overdue Days" column, reproduced exactly. In plain words: **how many days late (or early) did the shipment actually run against the supplier's promise** — with no grace period applied.

- **Signed:** shipped 7 days *before* the Estimated ETD → **−7**. A negative number means the supplier beat their promise.
- **Accrues daily while unshipped** — grows by one every day past the Estimated ETD.
- **Freezes at the shipment date** once Accounts records it.
- This is the day count that feeds Cost of Fund (Section 4).

4 Cost of Fund CLIENT RULE — 10-JUL-2026

Confirmed with the client on 10-Jul-2026. The 10-day grace period acts as a **threshold**: nothing is charged while the shipment stays within grace — but the moment grace is exceeded, the charge kicks in and counts **from the Estimated ETD itself** (day 1, not day 11).

```
While within grace (shipped, or waiting, up to Estimated ETD + 10 days):

    Cost of Fund = 0

Once the grace period is exceeded:

    Cost of Fund = Deposit Amount × 12% × ( (Actual Shipment Date or Today) − Estimated ETD )

Shipped EARLY (before the Estimated ETD):

    same formula — the day count is negative, giving a negative Cost of Fund (a notional gain
```

In plain words: **if the supplier stays inside the grace window nothing is charged; once they miss it, the charge is the interest the deposit money would have earned (at 12% per year) over the full delay measured from the promised ETD — and beating the ETD earns a notional gain.**

- **Within grace → zero.** ETD 7-Jul → grace runs until 17-Jul; shipping (or waiting) up to 17-Jul costs nothing.
- **Grace exceeded → retroactive.** Shipping on 18-Jul is charged for the full 11 days since 7-Jul, not just 1 day.
- **Shipped early → negative "notional gain"** (e.g. shipping 6 days before ETD shows a negative amount), exactly as in your sheet.
- **Not yet shipped → 0 until grace runs out, then accrues daily** from the Estimated ETD, so exposure stays visible and grows day by day.
- **Shipment date recorded → frozen.** The figure never changes again (see *When does the charge stop?* below).
- **No advance paid yet → 0.** No capital is locked, so no cost of fund.
- Rounded to 2 decimals. The 12% p.a. rate and the 10-day grace are admin-configurable, no code change needed.
- **Shown in the request's own currency.** A EUR request's Cost of Fund appears as €…, a CNY request's as ¥…, matching its deposit amount (since 10-Jul-2026). Company-wide "Total Cost of Fund" tiles still sum across all currencies — see the note in Section 6.

When does the charge stop? — recording the Ship Date

The charge (or the daily accrual on an unshipped order) stops the moment the **actual Ship Date** is entered into the system: the day count freezes at that date and the Cost of Fund never moves again. Because the payment is processed — and the record locked — long before the goods actually ship, the Ship Date is **deliberately kept open for entry and correction even after the record is locked**. It has its own dedicated box on the request's payment page.

- **Who can record it:** anyone from the Super Admin, Finance Admin or Accounts Team.
- **When:** at any time — before or after the payment is processed and locked. This is the only detail that stays editable on a locked record (alongside the TT copy).
- **Corrections:** allowed; every entry and change is written to the audit trail with the old and the new date, and by whom.
- **Effect is immediate:** the Cost of Fund recalculates the moment the date is saved — frozen at the recorded date, or shown as a notional gain if that date beat the Estimated ETD.

Worked examples (real rows from your own sheet)

Scenario	Deposit	Estimated ETD	Shipped	Days	Cost of Fund
Shipped late, past grace	49,059.00	16-Jan-2026	08-Feb-2026	+23	**370.97**
Shipped late but within grace	8,424.00	15-Jan-2026	25-Jan-2026	+10	**0.00**
Shipped early (notional gain)	5,832.00	15-Jan-2026	23-Dec-2025	−23	**−44.10**
Shipped exactly on ETD	7,446.13	31-Dec-2025	31-Dec-2025	0	**0.00**
Not yet shipped (as of 10-Jul-2026)	30,294.00	05-Jan-2026	—	+186	**1,852.50**

These rows — and more covering every scenario, including the grace boundary itself — are encoded as automated tests in the codebase, so any future code change that alters a result will fail the build.

> **One deliberate difference from your sheet.** Your sheet charges Cost of Fund even on shipments that arrive within the 10-day grace (e.g. it shows 27.70 on the 10-day row above). Per the rule confirmed on 10-Jul-2026, the system shows **0** for those rows. Everything else — late shipments past grace, early-shipment gains, unshipped accrual — matches your sheet to the cent (originally verified against all 534 rows).

> **Single source of truth.** Cost of Fund is computed in exactly one function and stored once per request. The Payment Queue snapshot, the HoM dashboard, all analytics tabs, charts and report exports read that same stored value — they cannot disagree. It recalculates immediately when a payment or shipment date is saved, when a request is edited, every 30 minutes automatically, and on the Recalculate button.

> **One thing to be aware of:** your sheet's *dashboard* tab contains a few aggregation quirks (summary ranges that stop partway down the data, fixed percentage denominators) that we deliberately did not copy. The system computes its totals from the same row-level values, so small differences against your sheet's dashboard percentages are the sheet's quirks — the row-level figures match exactly.

“Notional Gain / Loss” on dashboards

Everywhere a “Notional Gain” figure appears (overdue KPI tiles, delay buckets, by-merchandiser / vertical / customer tables) it is simply the **sum of Cost of Fund** over the rows in that group, split per currency. It is the same number under a different label, not a separate formula.

5 Payment Timing Metrics

Metric	Formula	Where shown
Payment-to-Ship Days	`Actual Shipment Date − Payment Date` (only when both exist)	Analytics table “Pmt→Ship”; Delay report; summary average
Payment-to-Request Days	`Payment Date − Request Created Date`	Analytics table “Pmt→Request”
Avg Payment-to-Ship	`AVERAGE(Payment-to-Ship Days)` over the filtered set	Analytics summary KPI

6 Dashboard & Analytics Aggregations

“Overdue” in every aggregate below means **Overdue Days > 0** (Section 3.2). Monetary sums are kept **per currency** (USD / CNY / EUR columns) — see Section 9.

> **Who sees these numbers (role scoping, since 11-Jul-2026).** Every analytics figure — summary KPIs, per-request table, monthly MoM/YTD trends and shipment KPIs — respects the viewer's role: **merchandisers see only their own requests** (submitted in-app or via the public form with their registered email); admin, finance and accounts roles see company-wide figures. Access to the specialised tabs (overdue by currency, delay buckets, by-merchandiser/vertical/customer) is additionally controlled per role from Admin → Analytics Access.

> **Cross-currency totals.** Per-request Cost of Fund is always displayed in the request's own currency. Single-figure tiles such as "Total Cost of Fund" sum ALL currencies together and are indicative only — the per-currency breakdowns (Section 6.2 buckets, overdue-by-currency KPIs) are the authoritative monetary views.

6.1 Summary KPIs

KPI	Definition
Total Requests	Count of all (non-deleted) requests in the filter
Pending Payment / Processed	Counts by current workflow status
Total Deposit Exposure	Sum of deposit amounts
Overdue Shipments	Count of requests with Overdue Days > 0
Total Cost of Fund	Sum of Cost of Fund (Section 4)

6.2 Delay buckets (aging analysis)

Buckets on Overdue Days: **Graced ETD** (0 or not overdue), then 15-day ranges — G-15, 15-30, 30-45, ... 135-150 — extended automatically in further 15-day steps if any request is more than 150 days overdue. A request falls in the bucket where `lower < Overdue Days ≤ upper`. Each bucket shows: case count, overdue deposit per currency, notional (Cost of Fund) per currency, and

```
% of Delayed Cases = bucket cases ÷ total overdue cases × 100      (1 decimal)
```

6.3 Breakdown tables (by Merchandiser / Vertical / Customer)

Each groups the *overdue* requests and shows: overdue cases, overdue deposit per currency, `Avg Delay = AVERAGE(Overdue Days)`, Notional Gain = sum of Cost of Fund. The **Contribution %** denominators differ by tab (flagged in the companion questionnaire):

Tab	Contribution % =
By Merchandiser	that merchandiser's overdue **USD amount** ÷ total overdue USD × 100
By Vertical	that vertical's overdue **case count** ÷ total overdue cases × 100
By Customer	that customer's overdue **case count** ÷ total overdue cases × 100

6.4 Monthly trends (MoM / YTD)

```
MoM (per month, by request created date):
  Total Overdue Days = SUM(Overdue Days)        Total Cost of Fund = SUM(Cost of Fund)

YTD = running cumulative total of the MoM values, January → December
```

6.5 Shipment KPIs

KPI	Definition
Total Active	All non-deleted requests
Shipped	Status = Payment Processed
Yet to Ship	Status = Pending Payment
Total Overdue	Overdue Days > 0
ETD −10 Grace Days	Requests approaching their deadline (within the 10-day window before Grace ETD)

6.6 NPA panel (HoM dashboard)

Overdue Suppliers list: every supplier with at least one overdue request; shows `Overdue Request Count` and `Max Overdue Days = MAX(Overdue Days)`, plus the flag details when the supplier is formally flagged.

Merchandiser performance: per merchandiser — total requests, overdue count, and `Avg Overdue Days = AVERAGE(Overdue Days, excluding zeros)`.

7 Supplier Auto-Flagging Rule

```
IF a request's Overdue Days > 0            (even 1 day past Grace ETD)
AND the supplier has no active default flag
THEN flag the supplier automatically:
     reason = "Auto-flagged: {N}d past ETD grace period"
     outstanding amount = 0 · flagged date = today · marked as auto-flagged
```

- One active flag per supplier — repeated overdue requests do not create duplicates.
- A flagged supplier's new requests are routed to **Head of Merchandiser approval** before entering the payment queue.
- Flags are cleared manually via **Resolve** on the Supplier Risk page.
- Cancelled and rejected requests can never trigger a flag.

8 Report Exports — Computed Columns

Report	Included rows	Computed columns
Request report	All requests in date range	Overdue Days, Cost of Fund (from stored snapshot — no new math)
Cost of Fund report	Only rows where Cost of Fund applies (advance paid + Estimated ETD present)	Cost of Fund (signed — early shipments appear as negative gains), **Delay Days** (= the grace-based Overdue Days of Section 3.2, shown for risk context; the Cost of Fund amount itself uses the grace-free day count of Section 3.4), sorted highest cost first
Delay report	Only overdue rows	Grace ETD, Overdue Days, Payment-to-Ship Days, Risk Status
Supplier / Payment reports	—	Pass-through of stored values, no calculations

Date-range filters are inclusive of the full end day.

9 What the System Deliberately Does NOT Calculate

- **No currency conversion.** USD, CNY and EUR figures are always aggregated in separate columns and never combined at a rate. The *Exchange Rate* field on a request is stored and shown in reports for reference only — it is not used in any formula.
- **Payment terms are descriptive.** The Balance Due / payment-terms selection (T/T, L/C, etc.) drives no calculation.
- **Deposit % is not enforced against the amounts.** On the in-app and public forms the percentage and the deposit amount are entered independently; the system does not check `amount = invoice × %` (only the Google Form channel auto-computes it).
- **Outstanding amount on auto-flags is always 0.** Only manual flags carry a finance-entered outstanding amount.

Advance Deposit Tracker — Formula & Calculation Reference · v2.3 · 11 July 2026 · Reflects production deployment of 11-Jul-2026 (Cost of Fund per the client-confirmed grace rule; Ship Date recordable post-lock by Super Admin / Finance Admin / Accounts Team to stop the charge; Cost of Fund displayed in each request's own currency; analytics role-scoped so merchandisers see only their own data; day counts and rate originally verified against the "Sunshine Deposits Analysis V2-PWA" sheet, 534 rows). Companion documents: “Advance Deposit Tracker — Engineering Case Study” (v1.2) and “Advance Deposit Tracker — Formula Confirmation Questionnaire” (v1.0).